

TSETSERLEG, Mongolia

WMO: 442820

Lat: 47.4717N

Lon: 101.4628E

Elev: 5548

StdP: 11.98

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-13.9	-28.8	1.3	-13.9	-25.4	1.6	-9.9	22.9	17.0	20.7	14.1	1.7	320	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	80.9	56.7	77.2	55.8	74.2	54.8	60.9	73.4	59.1	71.2	57.4	69.2	7.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.8	84.6	64.7	54.6	78.1	63.4	52.7	72.9	62.2	29.8	73.6	28.4	71.2	27.2	69.4	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	18.3	15.8	DB	-21.4	86.7	5.8	3.5	-25.6	89.2	-29.0	91.3	-32.3	93.2	-36.6	95.8
			WB	-21.8	64.5	5.8	3.0	-25.9	66.7	-29.3	68.5	-32.5	70.2	-36.7	72.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	5.0	10.9	23.3	38.3	48.5	58.0	61.8	58.6	49.0	35.1	20.1	9.7	(6)
(7)		DBStd	21.65	10.13	10.67	11.31	9.29	8.72	6.42	5.40	5.48	7.81	8.92	10.91	9.48	(7)
(8)		HDD50	6546	1395	1095	827	362	134	12	1	5	108	463	896	1248	(8)
(9)		HDD65	11004	1860	1515	1291	802	513	222	127	207	481	927	1346	1713	(9)
(10)		CDD50	1066	0	0	0	9	88	251	366	273	78	1	0	0	(10)
(11)		CDD65	50	0	0	0	0	1	11	28	10	0	0	0	0	(11)
(12)		CDH74	732	0	0	0	0	43	194	325	151	19	0	0	0	(12)
(13)		CDH80	139	0	0	0	0	3	36	74	25	1	0	0	0	(13)
(14)	Wind	WSAvg	5.5	4.5	5.0	6.3	7.3	7.0	5.5	4.4	4.7	5.3	5.5	5.6	5.0	(14)
(15)	Precipitation	PrecAvg	13.0	0.1	0.1	0.3	0.6	1.3	2.5	3.3	3.0	0.9	0.6	0.2	0.1	(15)
(16)		PrecMax	18.9	0.4	0.6	0.6	1.6	3.1	6.3	6.0	7.4	2.6	2.0	0.6	0.9	(16)
(17)		PrecMin	6.1	0.0	0.0	0.0	0.0	0.1	0.4	0.6	0.1	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.8	0.1	0.1	0.2	0.3	0.7	1.2	1.4	1.6	0.7	0.4	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	40.3	56.0	68.7	79.1	84.6	87.3	83.4	76.6	63.4	50.7	35.9	(19)
MCWB			26.4	29.2	38.5	46.1	51.2	56.8	60.4	57.8	52.9	43.3	36.5	27.0	(20)	
2%		DB	29.3	35.5	50.1	62.9	73.5	79.2	81.4	78.0	71.5	58.4	44.7	30.8	(21)	
		MCWB	22.3	26.3	35.2	42.8	49.9	54.7	58.7	56.5	50.7	41.4	33.2	23.5	(22)	
5%		DB	24.5	31.2	45.3	58.8	69.5	75.4	77.3	74.3	67.4	54.2	40.0	27.1	(23)	
		MCWB	18.6	23.5	32.3	41.1	48.0	53.7	57.8	55.5	48.7	39.4	30.2	20.8	(24)	
10%		DB	19.8	27.1	40.6	54.8	65.5	72.1	74.0	70.9	63.5	49.9	35.5	23.1	(25)	
		MCWB	15.0	20.7	29.7	38.9	46.2	53.1	56.8	54.7	47.0	37.0	27.4	17.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.8	29.6	39.3	47.0	55.7	60.4	65.3	62.9	55.7	45.0	37.6	27.3	(27)
MDCB			34.4	39.5	55.0	65.4	72.4	75.8	78.1	74.1	68.0	60.1	48.9	35.1	(28)	
2%		WB	22.2	26.4	35.4	44.0	51.7	58.0	61.9	59.9	52.9	42.5	33.5	23.6	(29)	
		MDCB	29.0	35.1	48.8	60.6	69.4	71.9	74.4	70.2	66.0	56.4	44.2	30.4	(30)	
5%		WB	18.6	23.6	32.6	41.7	49.2	56.3	60.0	58.0	50.7	40.0	30.5	20.8	(31)	
		MDCB	24.3	31.0	44.3	57.3	67.1	69.4	72.7	68.5	63.8	52.7	39.5	26.8	(32)	
10%		WB	15.1	20.7	30.0	39.3	47.0	54.7	58.4	56.4	48.4	37.5	27.7	17.8	(33)	
		MDCB	19.7	27.1	40.2	54.1	63.5	67.9	70.5	67.3	61.3	49.4	35.2	22.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.1	20.7	21.1	20.5	21.4	20.9	19.2	19.9	21.7	20.4	18.6	17.4	(35)	
5% DB		MCDBR	22.8	25.0	25.4	26.5	28.2	27.4	25.8	26.1	27.4	26.4	23.2	20.9	(36)	
		MCWBR	17.7	18.4	15.6	14.3	13.7	11.6	10.1	10.8	12.8	14.6	15.1	15.9	(37)	
		MCDBR	22.4	24.8	24.5	25.1	26.2	22.4	21.2	20.8	23.8	25.5	22.0	20.7	(38)	
(39)		5% WB	MCWBR	17.5	18.5	15.4	14.1	13.5	10.8	10.1	10.6	12.0	14.7	14.7	15.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.227	0.252	0.297	0.336	0.349	0.357	0.349	0.328	0.296	0.273	0.248	0.230	(40)	
(41)		taud	2.398	2.357	2.297	2.270	2.283	2.364	2.434	2.461	2.500	2.482	2.428	2.380	(41)	
(42)		Ebn at Noon	283	295	294	291	289	287	288	291	292	284	270	266	(42)	
(43)		Edh at Noon	21	27	34	39	40	37	35	32	28	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	611	949	1419	1802	2008	1976	1874	1694	1458	1002	638	490	(44)	
(45)		RadStd	21	32	49	78	134	110	114	86	71	51	34	16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	-4.79	N/A	-1.21	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	N/A	N/A	-4.66	N/A	-1.40	-1.80	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page